# PRATYUSH AVI

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## EDUCATION

**New York University** September 2023 – May 2028 (expected)
*PhD, Computer Science* *New York, NY*

*Prospective Thesis Focus*: Applying techniques from theoretical computer science and random numerical linear algebra for practically faster and provably efficient algorithms for problems in scientific machine learning and approximate nearest neighbor search.

I am currently supported by the Department of Education GAANN Fellowship.

**Arizona State University** August 2019 – December 2022
*BSE, Computer Systems Engineering & Barrett, The Honors College* *Tempe, AZ*
**GPA:** 4.00

## PUBLICATIONS

**Almost Navigable Graphs**
**P. Avi**, Christopher Musco
***Best Paper*** *at The 2nd Workshop on Vector Databases (VecDB)* at VLDB 2026, arXiv:2607.14564

**Query Efficient Structured Matrix Learning**
Noah Amsel, **P. Avi**, Tyler Chen, Feyza Duman Keles, Chinmay Hegde, Christopher Musco, Cameron Musco, David Persson
*Conference on Learning Theory 2026*, arXiv:2507.19290
Preliminary version at *Theory of Scientific Computing Workshop* at COLT 2025
Poster presentations at COLT 2025 and Simons Institute Workshop on Linear Systems and Eigenvalue Problems 2025

## EXPERIENCE

**PhD Research Intern** May 2023 – August 2023
*MIT Lincoln Laboratory, Group 33 - Advanced Sensor Systems & Test Beds* *Lexington, MA*

- Built a real-time planning web interface using Python and React.js, improving user interface usability and deployment for mission operators.
- Engineered a domain-specific planning language to automate real-time mission execution and reduce operator workload.
- Mentored junior interns, providing technical guidance to support project completion.

**Undergraduate Research Intern** May 2022 – August 2022
*MIT Lincoln Laboratory, Group 33 - Advanced Sensor Systems & Test Beds* *Lexington, MA*

- Built scalable machine learning and signal processing models (CUDA, PyTorch) to detect key events in large-scale radar datasets.
- Implemented YOLOv3 for spectrogram-based signal classification and prototyped a graph neural network model that used raw-spectrogram data for signal recognition.
- Developed a user interface in React.js for a secure meeting recording and trancription utility.
- Gained hands-on radar experience through field-deployment project.

**Machine Learning Undergraduate Researcher** May 2021 – April 2022
*Purdue University & U.S. Army* *West Lafayette, IN*

- Built a graph neural network in PyTorch to predict properties of organic molecules relevant to energetics and material science research.
- Wrote a Jupyter Notebook based teaching utility to teach machine learning approaches to undergraduate students.
- Paper Link: Machine Learning for Data Driven Energetics Materials Screening and Performance Design
- Presented findings to leading energetics and Army Research Lab researchers.

## Projects

**Vector-Quantizer Web Playground** | *Artificial Intelligence, Data Science*

- Built a web-based playground for composing and testing different vector quantization methods, based on the VQ-Bench library: VQ-Playground

**Dimensionality Reduction for Spatial Audio** | *Artificial Intelligence, Data Science, Extended Reality*

- Used random binary matrices for compressing Head-Related-Transfer-Functions for spatial audio
  Project Report Link: Random Projections for Dimensionality Reduction of HRTFs

**Evaluating Hierarchical Learning using MAXQ and Q-Learning** | *Artificial Intelligence, Reinforcement Learning*

- Re-implemented and evaluated hierarchical learning by comparing MAXQ and Q-Learning in the OpenAI Gym.
  Project Report Link: Re-implementation of MAXQ with the OpenAI Gym in the Taxi Domain

## Teaching

**Teaching Assistant** September 2026 – Present 2026
*New York University* *New York, NY*

- **Course:** Algorithmic Machine Learning and Data Science with Prof. Ainesh Bakshi
- Course Website

**Teaching Assistant** January 2026 – May 2026
*New York University* *New York, NY*

- **Course:** Algorithmic Machine Learning and Data Science with Prof. Ainesh Bakshi
- Course Website

**Teaching Assistant** January 2025 – May 2025
*New York University* *New York, NY*

- **Course:** Algorithmic Machine Learning and Data Science with Prof. Chris Musco
- Course Website

**Teaching Assistant** January 2021 – December 2022
*Arizona State University* *Tempe, AZ*

- **Course:** Introduction to Programming Languages with Prof. Justin Selgrad

## Awards & Honors

**Best Paper** September 2026
*Workshop on Vector Databases (VecDB) at VLDB* *Boston, MA*

- Awarded best paper for work on *Almost Navigable Graphs*. Was one of only six papers invited for an oral presentation.

**Best Poster** July 2022
*MIT Lincoln Laboratory* *Lexington, MA*

- Led finalist team at MIT LL's Intern Innovation Idea Challenge, pitching a novel deep-sea exploration vehicle concept.
- Conceptualized vehicle design using MIT LL technologies, along with a supporting research and development pipeline.

**Dean's List** August 2019 - December 2022
*Arizona State University* *Tempe, AZ*

- All semesters through Fall 2019 - Fall 2022

## Technical Skills

**Languages:** Python, C/C++, Java, Dart, JavaScript, CSS/SASS, HTML, Julia, Verilog, RISC-V, MySQL
**Frameworks:** AWS, PyTorch, Tensorflow, Jupyter, CUDA, Vue.js, d3.js, Flutter, React.js, Electron.js
**Developer Tools:** Linux/UNIX, Intel Pin, Arduino, Raspberry Pi, Git, Firebase, MATLAB